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WaveSpace: analysis and simulation of cortical traveling waves

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Summary

Oscillatory cortical activity has been found to systematically propagate across space as traveling waves (Muller et al., 2018). Growing evidence links those spatiotemporal patterns to cognitive functions including, but not limited to visual spatial attention (Fakche & Dugué, 2024), memory (Mohan et al., 2024) and consciousness (Bhattacharya et al., 2022). Detecting and characterizing traveling waves in non-invasive multichannel recordings of brain activity requires multiple processing and analysis steps, some of which are interchangeable while others can lead to diverging interpretations of the same data. WaveSpace is a modular Python toolbox for the simulation and analysis of traveling wave dynamics in multichannel recording arrays. Based on a single, flexible data class users can execute, compare and recombine multiple commonly used analysis steps as well as evaluate their performance against simulated benchmarks.

Statement of need

Various approaches to detect and characterize cortical traveling waves have emerged in the literature. Typically, laboratories develop customized pipelines tailored to their experimental requirements and software platform preferences (Alamia & VanRullen, 2019; Alexander et al., 2006; Das et al., 2022; Muller et al., 2014; but see also Gutzen et al., 2024 for a notable exception).

The diversity of methods and implementations found in the literature poses challenges for researchers, both in selecting the one most suitable for their own studies and in directly comparing the performance of different pipelines. WaveSpace addresses this gap by integrating commonly used strategies into a single modular framework. This framework ensures that modules for preprocessing, data decomposition, spatial arrangement of sensor positions, wave analysis, and evaluation are interchangeable within the same workflow. Additionally, a simulation module allows for the generation of benchmarking data with desired properties to directly test the accuracy and specificity of planned analysis pipelines in silico. The resulting pipelines are ready-to-use in empirical studies (Fakche et al., 2024; Petras et al., 2025).

State of the field

Several single purpose pipelines for the analysis of cortical traveling waves exist. Table 1 shows a non-exhaustive list of openly available analysis code. In most cases, the provided code accompanies, and is tailored towards, a single experimental contribution to the scientific literature on cortical traveling waves.

Table 1. Single-purpose tools for traveling-wave analysis.

Tool	Lang	Method	Repo
wave-matlab	MATLAB	Phase–distance correlation	GitHub
Travelling wave analysis (ScaleSymmetry)	Python	SVD phase waves	GitHub
travellingWaveEEG	MATLAB	2D FFT	GitHub
Travelling wave analysis (jacobslab)	MATLAB	Circular–linear correlation	GitHub
NeuroPattToolbox	MATLAB	Optical flow	GitHub
cobrawap	Python	Optical flow (Snakemake)	GitHub
Phase vs Granger	MATLAB	Phase gradient + Granger	GitHub

In principle, multi-purpose neurophysiology data analysis packages such as fieldtrip (Oostenveld et al., 2011) for Matlab or MNE (Gramfort et al., 2014) for Python could be extended to include traveling wave analysis methods. Given the wide variety of available approaches and the lack of systematic comparison in the literature, WaveSpace was instead designed as a standalone tool with consistent workflows dedicated exclusively to traveling wave analysis. However, to ensure users can still benefit from the pre-processing, time-frequency decomposition and visualization methods provided by MNE, WaveSpace easily integrates MNE data objects at any stage of processing.

Functionality and software design

WaveSpace has been developed to provide an array of methods for the detection and analysis of cortical traveling waves, primarily in non-invasive electrophysiology data such as electro- or magnetoencephalography. All WaveSpace functionality is based on a single data class, called WaveData, that enforces conventions for data dimension order and dimension naming. This allows for most processing steps to be interchangeable. The WaveData class organizes the in- and output of consecutive processing steps into discrete data-buckets, while logging progress. The entire framework is comprehensively documented and includes example scripts to facilitate adoption.

WaveSpace contains 6 core modules (see figure 1 for module overview):

- Decomposition: Provides multiple techniques to decompose broadband data into frequency components, including FFT-based methods (e.g., wavelets, filter-Hilbert), empirical mode decomposition (EMD), and generalized phase analysis.
- Spatial Arrangement: Includes methods to map 3D sensor positions onto 2D regular grids using approaches such as multidimensional scaling (MDS) and isomap. Multiple interpolation options are available.
- Wave Analysis: Offers a variety of analysis methods, such as 2D FFT, optical flow analysis, phase gradient methods, and principal component analysis (PCA).
- Simulation: Functions to simulate traveling and spatially stationary (i.e., standing) waves with both linear and nonlinear properties, as well as incorporate noise.
- Evaluation and statistics: Summarizes wave characteristics. Contains options for wave-scoring.
- Plotting: Contains visualization tools for each analysis option.

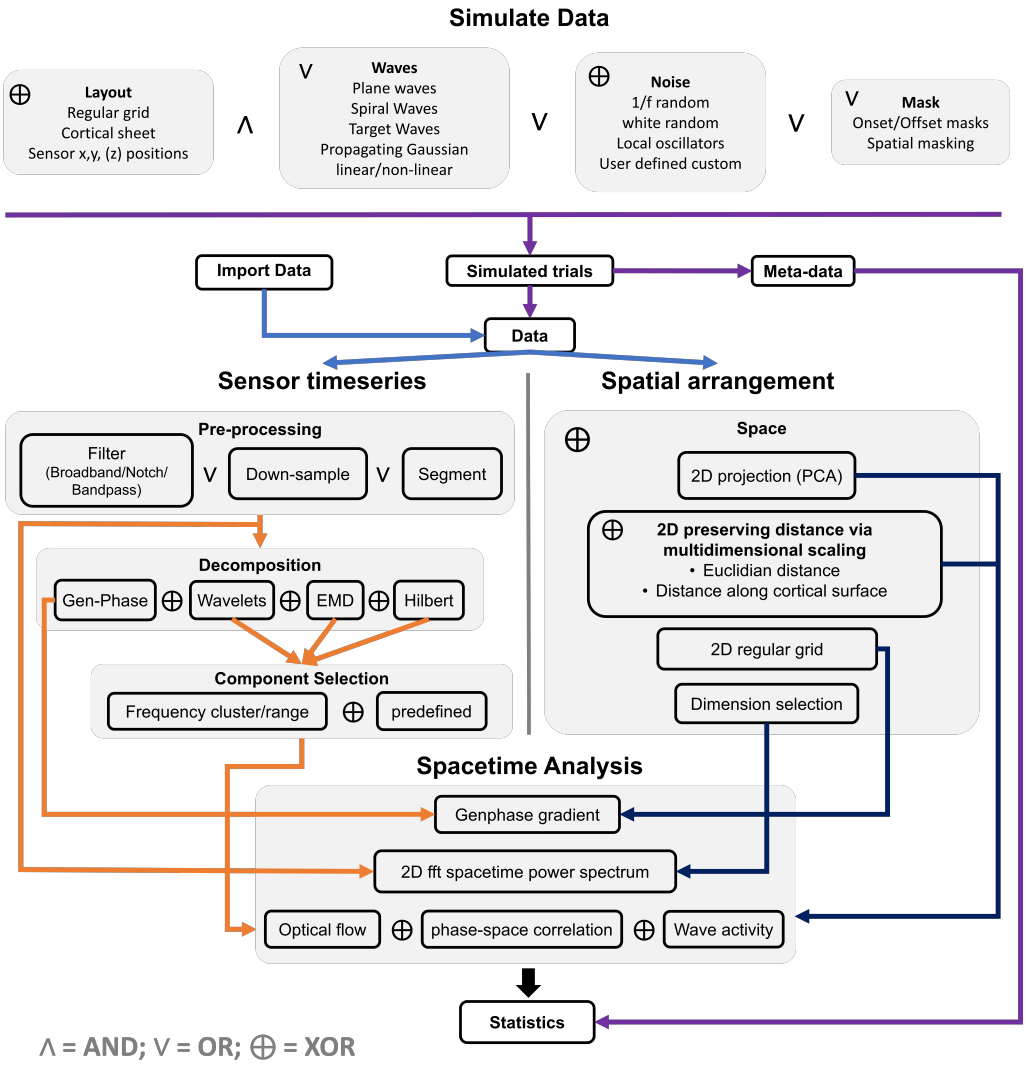


Figure 1: Overview of WaveSpace modules.

Research impact statement

WaveSpace has been publicly introduced during a workshop at the 47th European conference on visual perception (2025) and used in peer reviewed as well as ongoing work (Fakche et al., 2024; Kong et al., 2025; Petras et al., 2025). Users are invited to contribute to the ongoing package development via github.

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Toolbox dependencies

Environment file

AI usage disclosure

Github copilot in "ask" mode has been used in the initial translation of Matlab code to Python. When the resulting code was found to be inaccurate and failed to match the style of the rest of the package, most of it was manually re-written. Single word autocomplete was used throughout for code and comments. Copilot was used for code and formatting suggestions. Agentic AI was used to draft documentation.

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